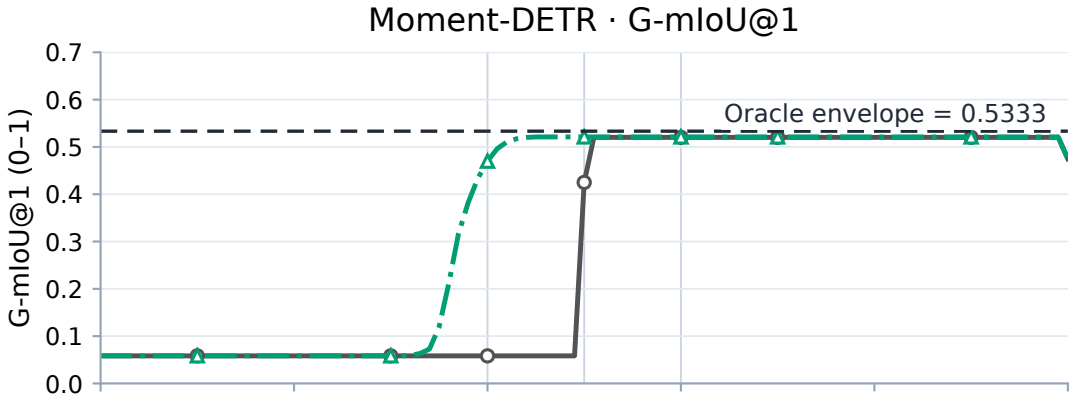
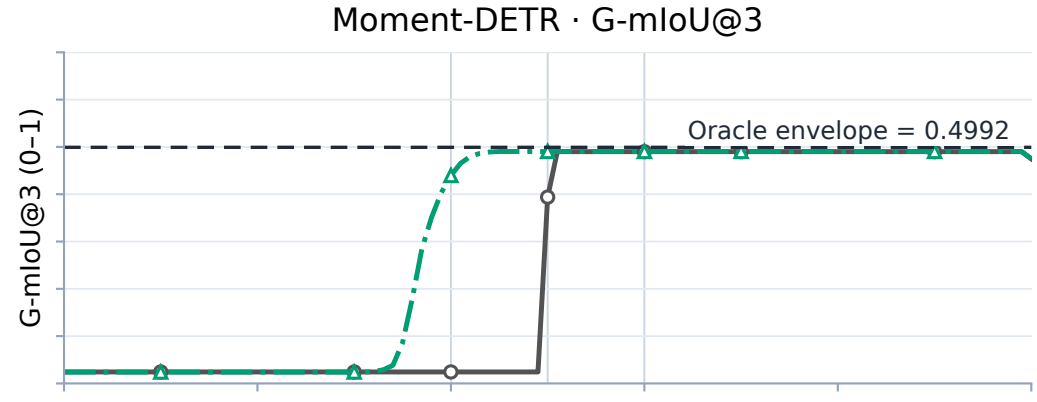
G-mIoU sensitivity by backbone · mAP and changes versus GMR

Standard-trained, fixed Val-selected checkpoints → SoccerGMR Standard Test (1036 queries)

Historical GEQ protocol: 6 non-improving epochs; no Test selection or tuning

Moment-DETR

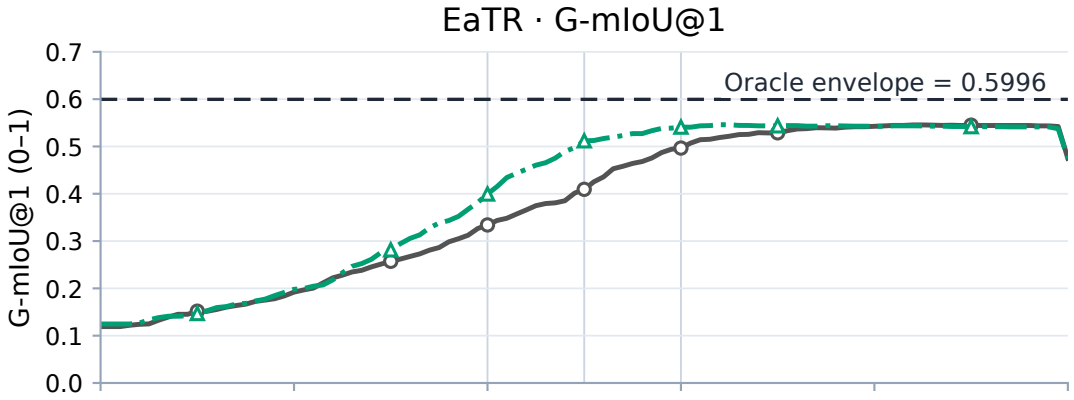


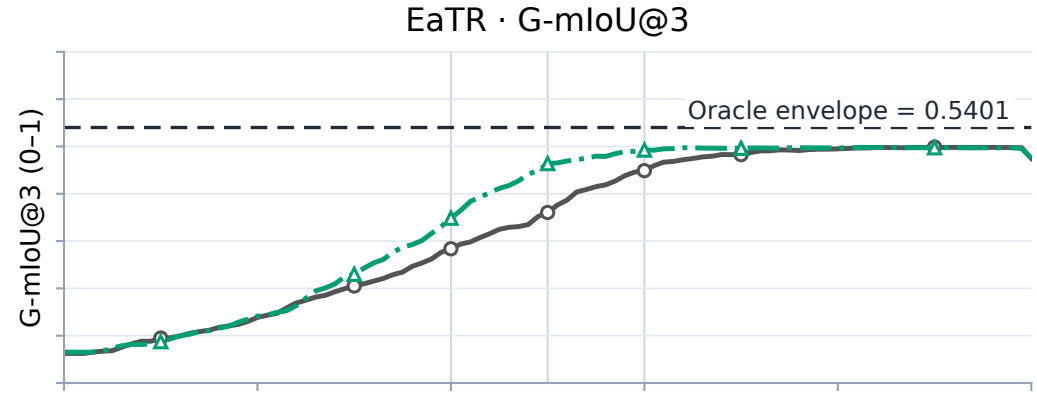


GMR baseline | mAP 6.68% — GMR + GEQ Full | mAP 5.77%

ΔmAP vs GMR: -0.91 pp · ΔG1 mean (.4/.5/.6): +16.95 pp · ΔG3 mean (.4/.5/.6): +17.06 pp

EaTR

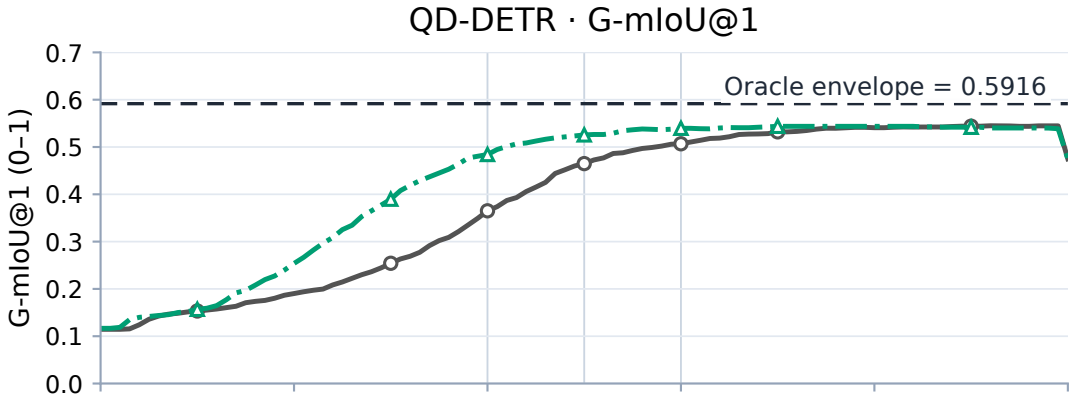


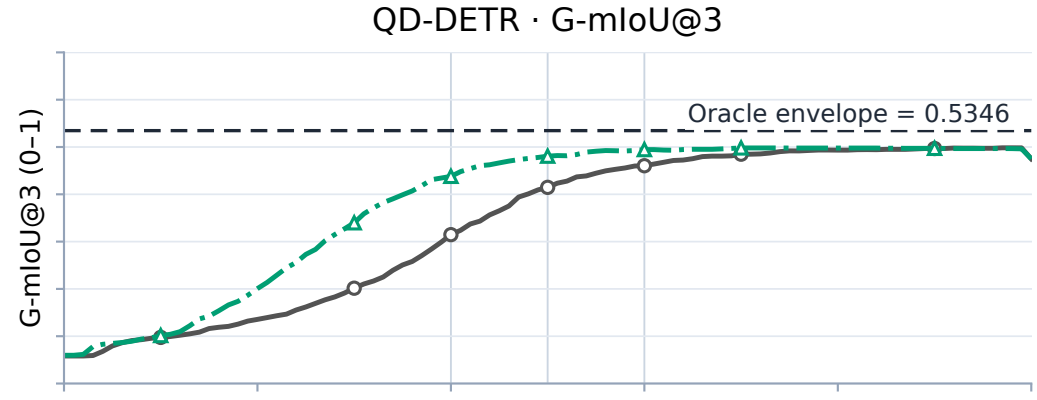


GMR baseline | mAP 16.49% — GMR + GEQ Full | mAP 17.48%

ΔmAP vs GMR: +0.99 pp · ΔG1 mean (.4/.5/.6): +7.07 pp · ΔG3 mean (.4/.5/.6): +7.05 pp

QD-DETR

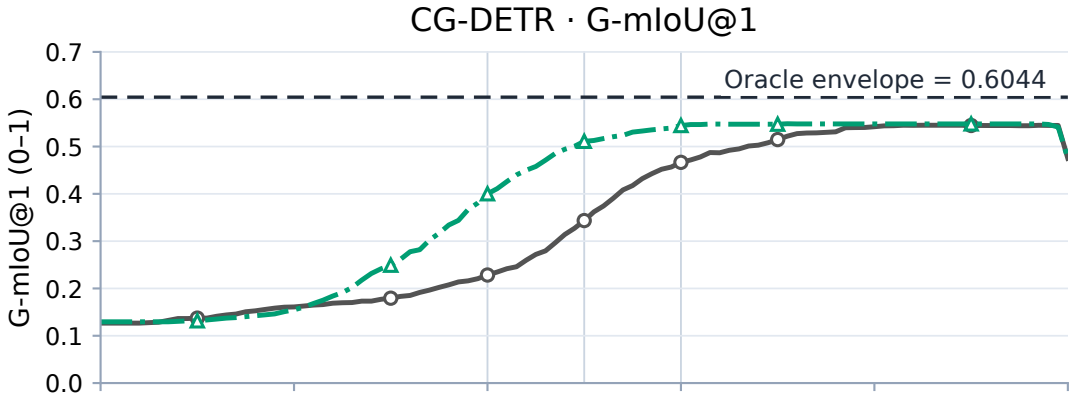


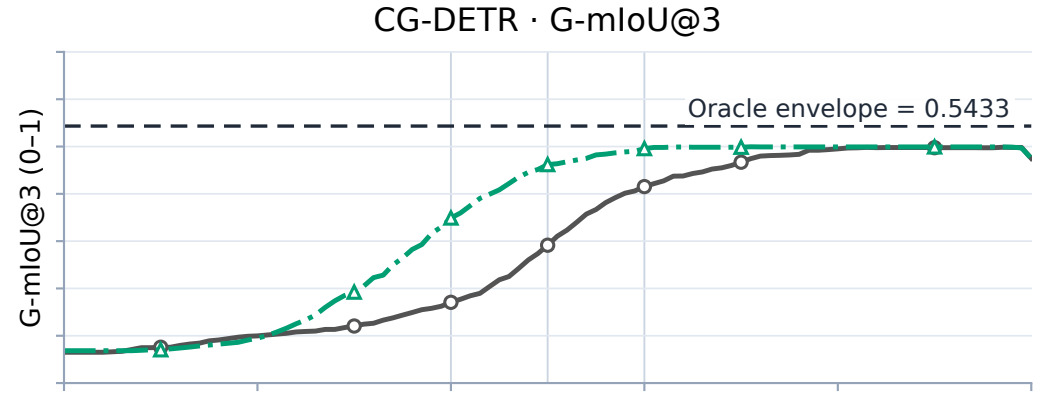


GMR baseline | mAP 14.62% — GMR + GEQ Full | mAP 14.83%

ΔmAP vs GMR: +0.21 pp · ΔG1 mean (.4/.5/.6): +7.08 pp · ΔG3 mean (.4/.5/.6): +7.44 pp

CG-DETR

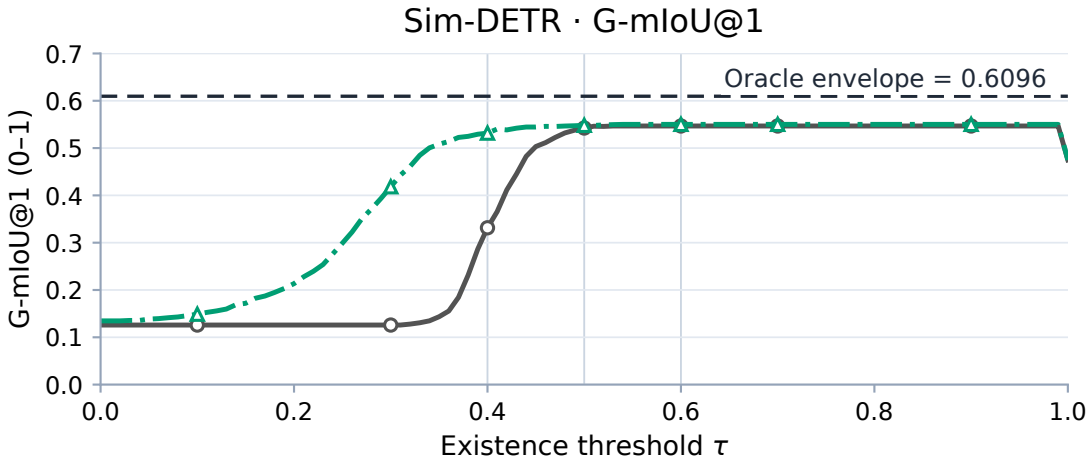


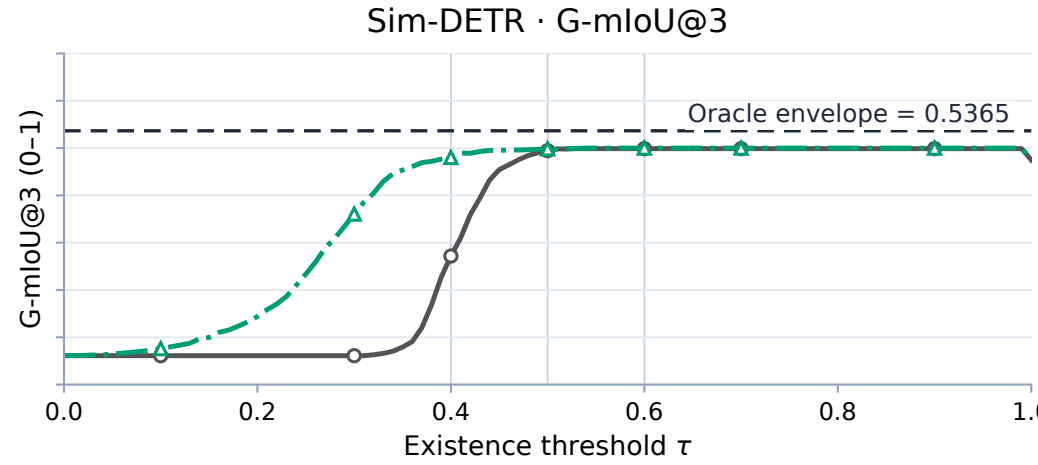


GMR baseline | mAP 18.65% — GMR + GEQ Full | mAP 18.71%

ΔmAP vs GMR: +0.06 pp · ΔG1 mean (.4/.5/.6): +13.90 pp · ΔG3 mean (.4/.5/.6): +14.30 pp

Sim-DETR





GMR baseline | mAP 20.73% — GMR + GEQ Full | mAP 22.26%

ΔmAP vs GMR: +1.53 pp · ΔG1 mean (.4/.5/.6): +6.97 pp · ΔG3 mean (.4/.5/.6): +7.11 pp

Oracle envelope: max of the two plotted methods with GT presence and each method's fixed candidates.

Curves: all 101 fixed thresholds. Δ: GEQ Full − matching GMR. G1/G3 means use τ = .4/.5/.6; pp = percentage points.

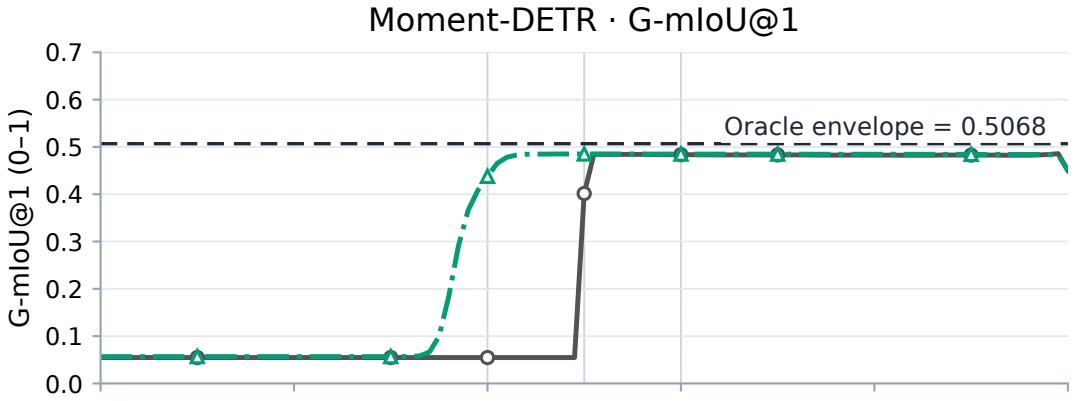
One fixed checkpoint per method and backbone; lines show threshold sensitivity, not repeated-run uncertainty.

G-mIoU sensitivity by backbone · mAP and changes versus GMR

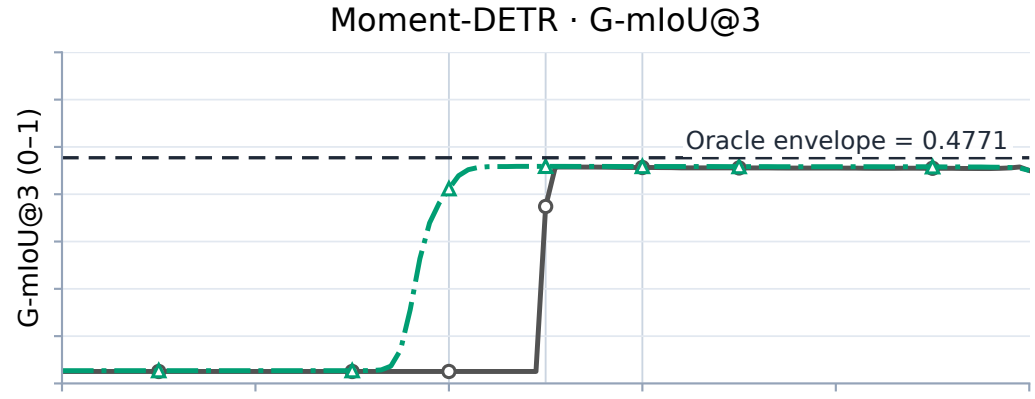
Standard-trained, fixed Val-selected checkpoints → SoccerGMR Full-Decodable Test (2970/2986 queries)

Historical GEQ protocol: 6 non-improving epochs; no Test selection or tuning

Moment-DETR · G-mIoU@1



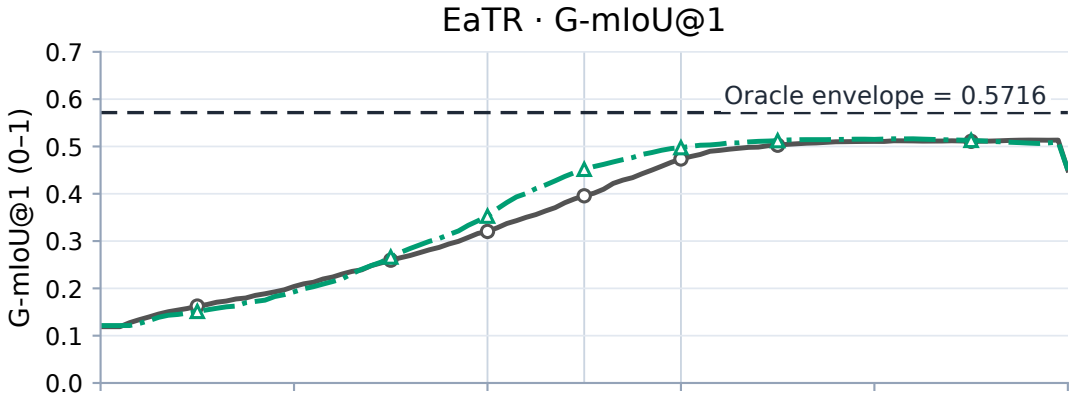
Moment-DETR · G-mIoU@3



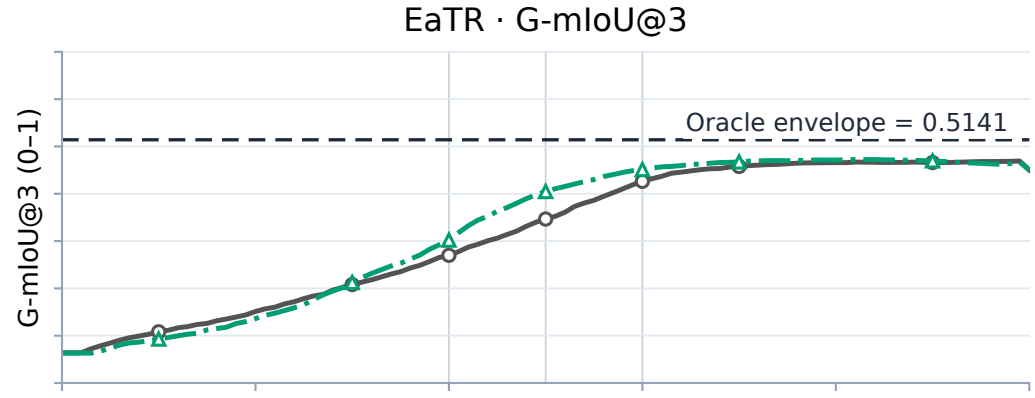
GMR baseline | mAP 5.47% — GMR + GEQ Full | mAP 5.47%

ΔmAP vs GMR: +0.00 pp — ΔG1 mean (.4/.5/.6): +15.60 pp — ΔG3 mean (.4/.5/.6): +15.77 pp

EaTR · G-mIoU@1



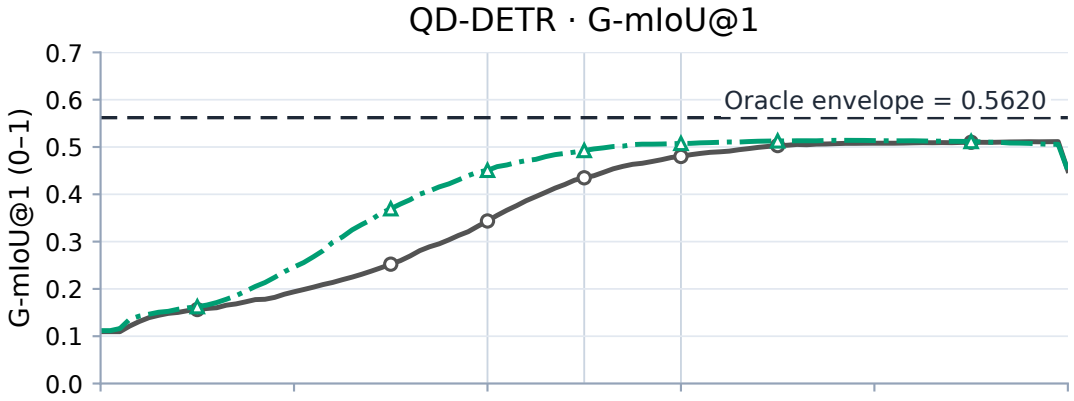
EaTR · G-mIoU@3



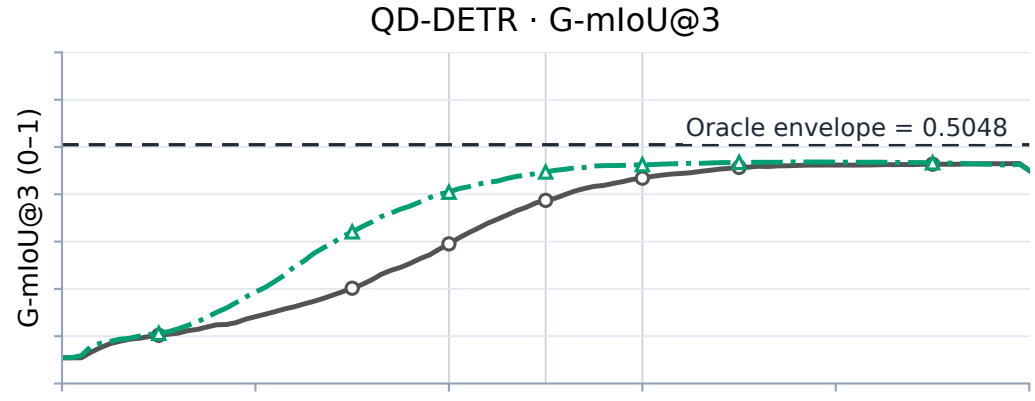
GMR baseline | mAP 15.19% — GMR + GEQ Full | mAP 15.26%

ΔmAP vs GMR: +0.07 pp — ΔG1 mean (.4/.5/.6): +3.73 pp — ΔG3 mean (.4/.5/.6): +3.84 pp

QD-DETR · G-mIoU@1



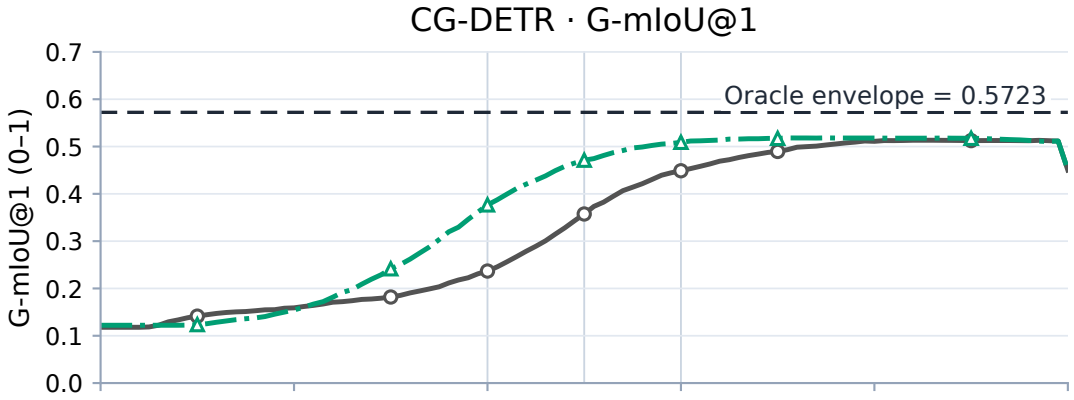
QD-DETR · G-mIoU@3



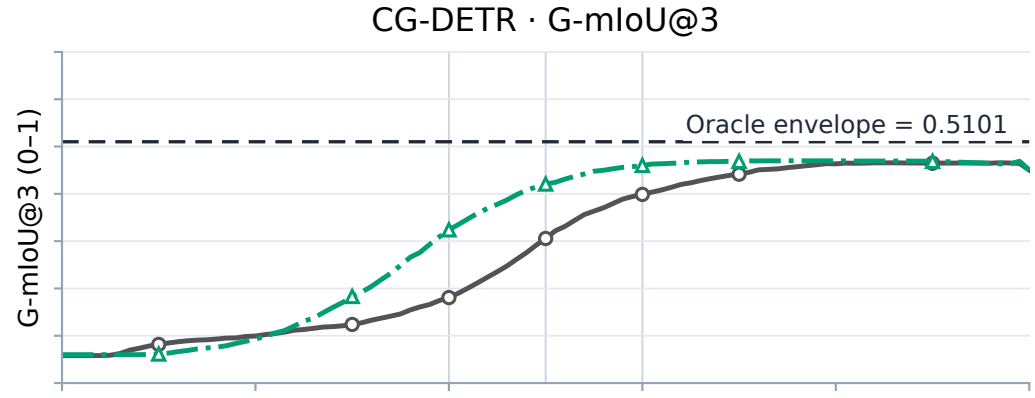
GMR baseline | mAP 12.65% — GMR + GEQ Full | mAP 13.22%

ΔmAP vs GMR: +0.57 pp — ΔG1 mean (.4/.5/.6): +6.40 pp — ΔG3 mean (.4/.5/.6): +6.59 pp

CG-DETR · G-mIoU@1



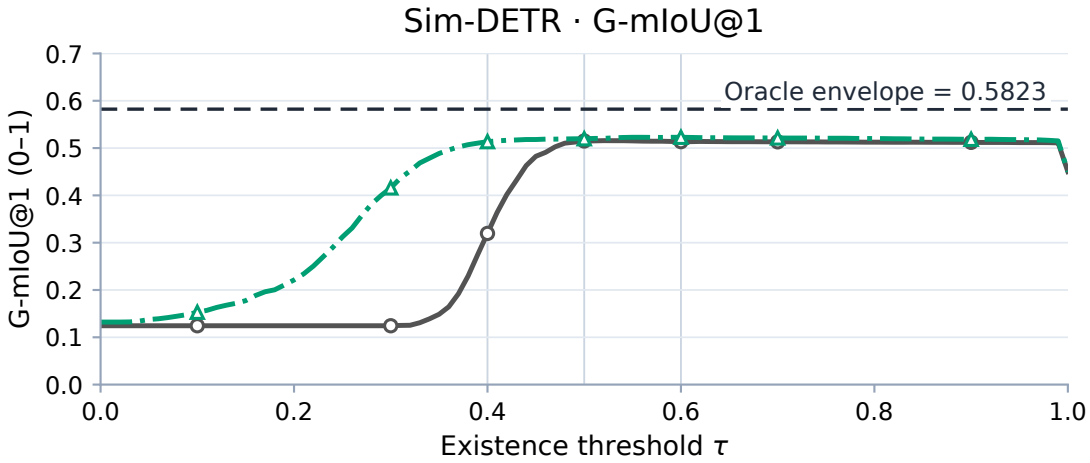
CG-DETR · G-mIoU@3



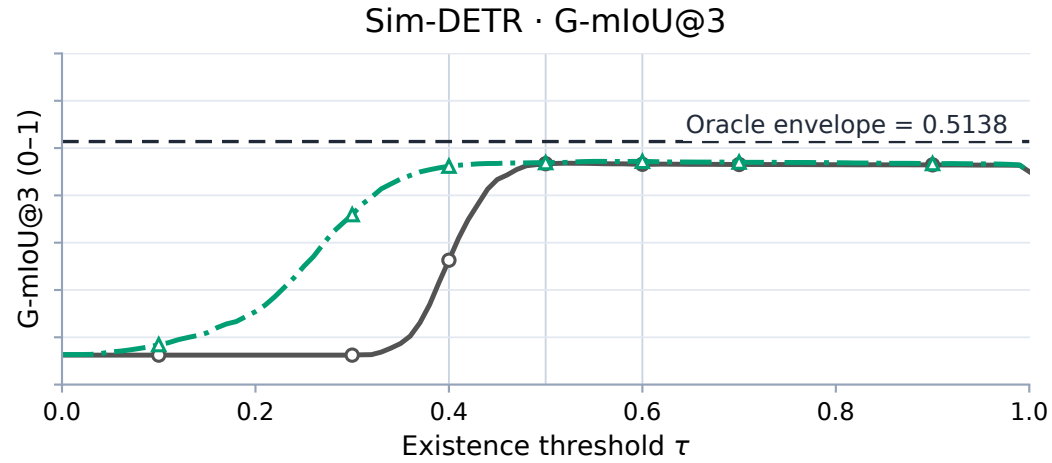
GMR baseline | mAP 14.67% — GMR + GEQ Full | mAP 14.85%

ΔmAP vs GMR: +0.18 pp — ΔG1 mean (.4/.5/.6): +10.45 pp — ΔG3 mean (.4/.5/.6): +10.65 pp

Sim-DETR · G-mIoU@1



Sim-DETR · G-mIoU@3



GMR baseline | mAP 19.18% — GMR + GEQ Full | mAP 20.58%

ΔmAP vs GMR: +1.40 pp — ΔG1 mean (.4/.5/.6): +6.93 pp — ΔG3 mean (.4/.5/.6): +6.90 pp

Oracle envelope: max of the two plotted methods with GT presence and each method's fixed candidates.

Curves: all 101 fixed thresholds. Δ: GEQ Full − matching GMR. G1/G3 means use τ = .4/.5/.6; pp = percentage points.

Legacy 75-token input limit retained: two long Full videos are truncated; 16 undecodable queries remain uncovered.